PART A
Experiment 2

A.1 Aim

Implement Tracking Sprint Progress using Burndown and Velocity Charts in Jira.

A.2 Prerequisite

- Basic knowledge of Agile Methodology and Scrum Framework
- Understanding of Sprint Planning and Sprint Execution
- Familiarity with Product Backlog and Sprint Backlog
- Basic knowledge of Jira Software
- Understanding of Story Points and Agile Estimation

A.3 Outcome

After successful completion of this experiment, students will be able to:

- Create and manage Sprint Backlogs in Jira.
- Monitor Sprint progress using Burndown Charts.
- Analyze team performance using Velocity Charts.
- Track completed and remaining work during a Sprint.
- Evaluate Sprint efficiency and team productivity.
- Generate Agile reports for project monitoring.

A.4 Theory

1. What is Tracking Sprint Progress using Burndown and Velocity Charts in Jira?

Tracking Sprint progress is an important activity in Agile project management. It helps the development team and project manager monitor the amount of work completed, the work remaining, and the overall performance of the team during a Sprint.

In Jira, Sprint progress can be monitored using reports such as the **Sprint Burndown Chart** and **Velocity Chart**.

A **Burndown Chart** shows the amount of work remaining in a Sprint over time. The horizontal axis represents the Sprint duration or dates, while the vertical axis represents the remaining work, such as Story Points or issues. As team members complete their assigned tasks, the remaining work should decrease. This helps the team determine whether the Sprint is progressing according to the planned schedule.

A **Velocity Chart** shows the amount of work committed to and completed by the team in completed Sprints. It helps measure the team's delivery capacity and can be used to estimate how much work the team may be able to complete in future Sprints.

Therefore, Burndown and Velocity Charts provide a visual method for monitoring Sprint progress, identifying delays or risks, and improving future Sprint planning.

2.Explain the Application in Detail (explain)

In this experiment, **Jira Software** is used to manage and monitor an Agile project named **Transport_management_system**.

The project is organized using Scrum concepts such as work items, backlog, Sprint, and workflow stages. Different work items were created for the project, including requirements analysis, design wireframe, web development, data storage, vehicle management, vehicle booking, and other project-related activities.

The work items are managed through the Jira board using the following workflow stages:

To Do → In Progress → In Review → Done

The experiment involves creating and managing a Sprint and monitoring its progress using Jira reports.

The **Sprint Burndown Chart** is used to observe the remaining work during the active Sprint. The chart displays how the work should decrease as tasks or work items are completed.

The **Velocity Report** is used to analyze the amount of work completed after a Sprint is completed. In this experiment, the report initially showed zero or no significant data because there was limited completed Sprint work and no completed work associated with Story Points. This demonstrates that meaningful Velocity data is generated after completing Sprints and recording completed work.

Thus, Jira provides a centralized platform for planning, tracking, monitoring, and analyzing Agile project activities.

3.Which Agile Framework, Tool, and Features are Used? (explain)

Agile Framework Used: Scrum

The Agile framework used in this experiment is **Scrum**. Scrum divides the project into smaller time-based iterations called **Sprints**. During each Sprint, the team selects work from the Product Backlog and attempts to complete the planned work.

The main Scrum components used are:

- Product Backlog
- Sprint Planning
- Sprint Backlog
- Active Sprint
- Daily Scrum
- Sprint Review
- Sprint Retrospective

Tool Used: Jira Software

Jira Software is used as the Agile project management tool. It helps teams organize project requirements, create work items, manage Sprints, track progress, and generate reports.

Jira Features Used

The following Jira features are used in this experiment:

1. **Backlog** – Used to manage and organize project requirements and work items.
2. **Work Items/Issues** – Used to represent tasks, stories, features, and other project activities.
3. **Sprint** – Used to plan and execute a fixed period of work.
4. **Scrum Board** – Used to visualize the current status of work items.
5. **Workflow Columns** – Used to track work through stages such as To Do, In Progress, In Review, and Done.
6. **Reports** – Used to analyze project and Sprint performance.
7. **Sprint Burndown Chart** – Used to monitor the remaining work during a Sprint.
8. **Velocity Report** – Used to analyze committed and completed work across completed Sprints.
9. **Work Item Details** – Used to view information such as work type, priority, assignee, and summary.

4. Common Workflow

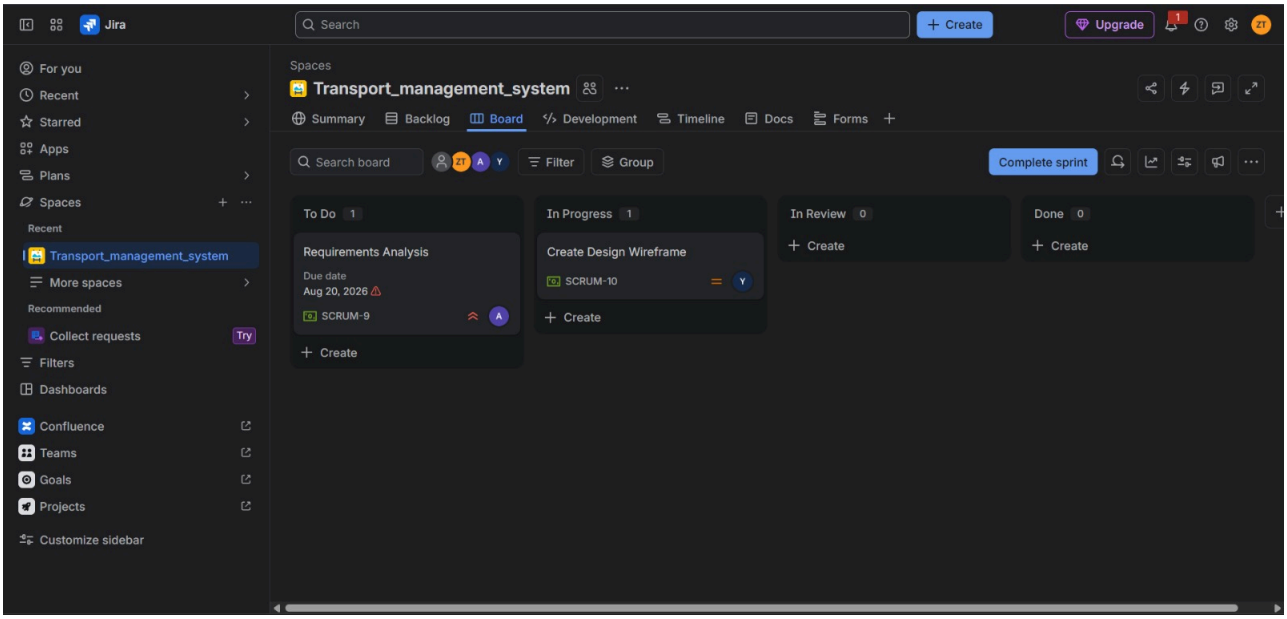
```
Product Backlog
        ↓
Sprint Planning
        ↓
Sprint Backlog
        ↓
Active Sprint
        ↓
Daily Scrum
        ↓
Burndown Monitoring
        ↓
Velocity Analysis
        ↓
Sprint Review
        ↓
Sprint Retrospective
        ↓
Next Sprint
```

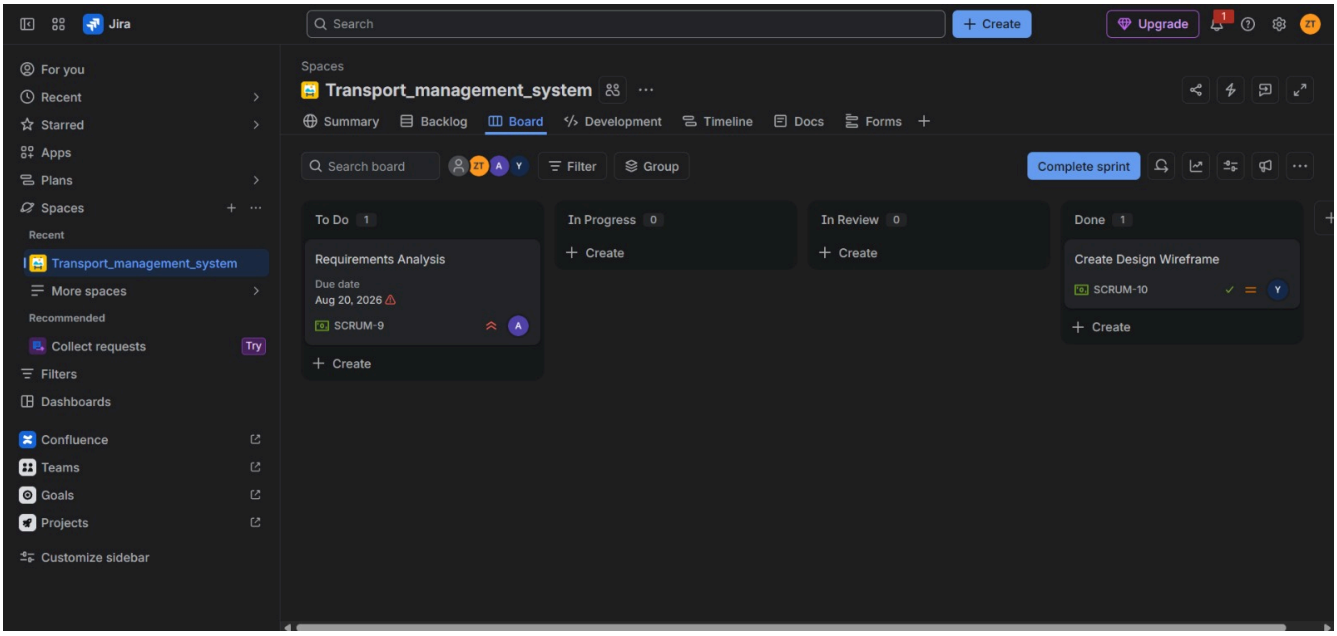
This experiment enables students to understand how Agile teams monitor Sprint execution using Jira's Burndown and Velocity Charts, evaluate team performance, identify project risks, and improve Sprint planning through continuous tracking and data-driven decision-making.

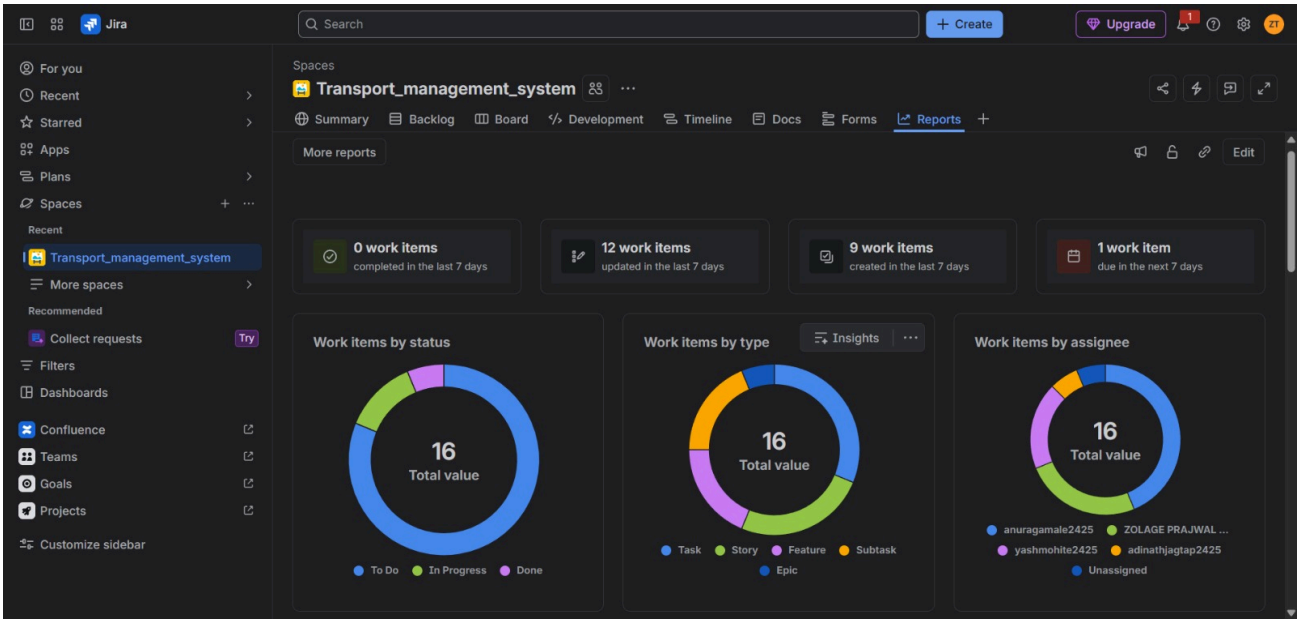
PART B
(PART B: TO BE COMPLETED BY STUDENTS)

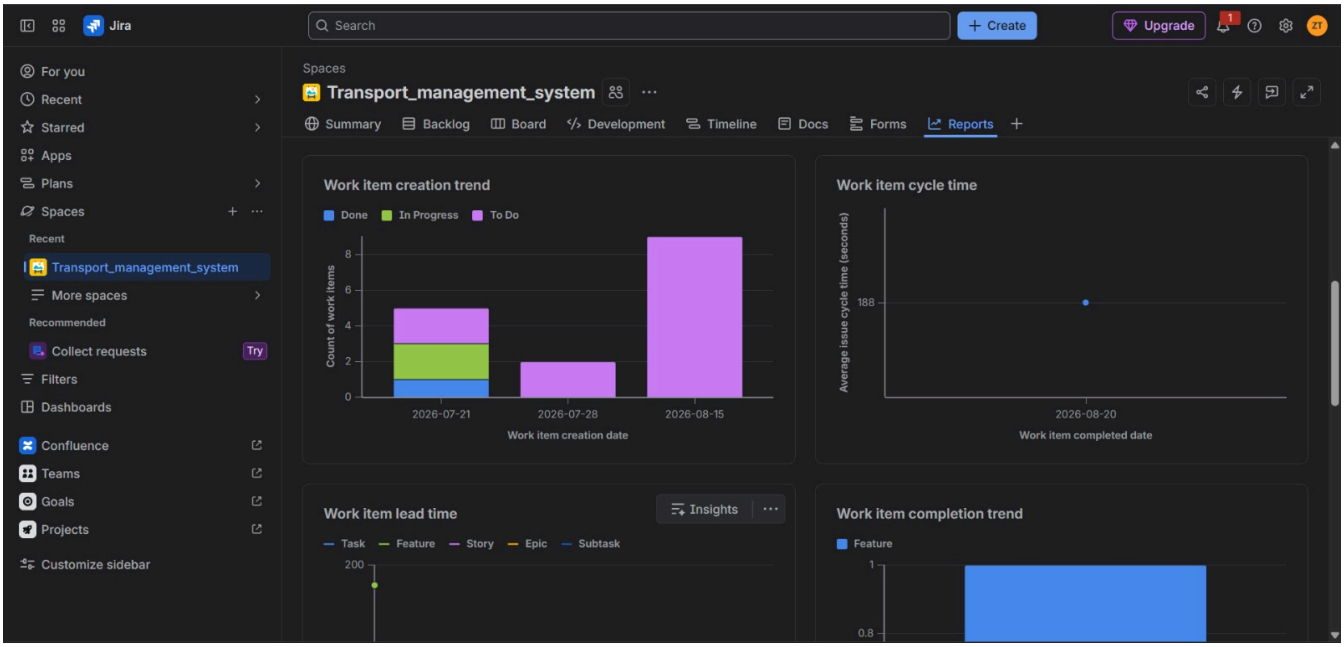
Roll No. A27	Name: Prajwal Ashok Zolage
Class: BE–AI&DS	Batch:A1
Date of Experiment:	Date of Submission:

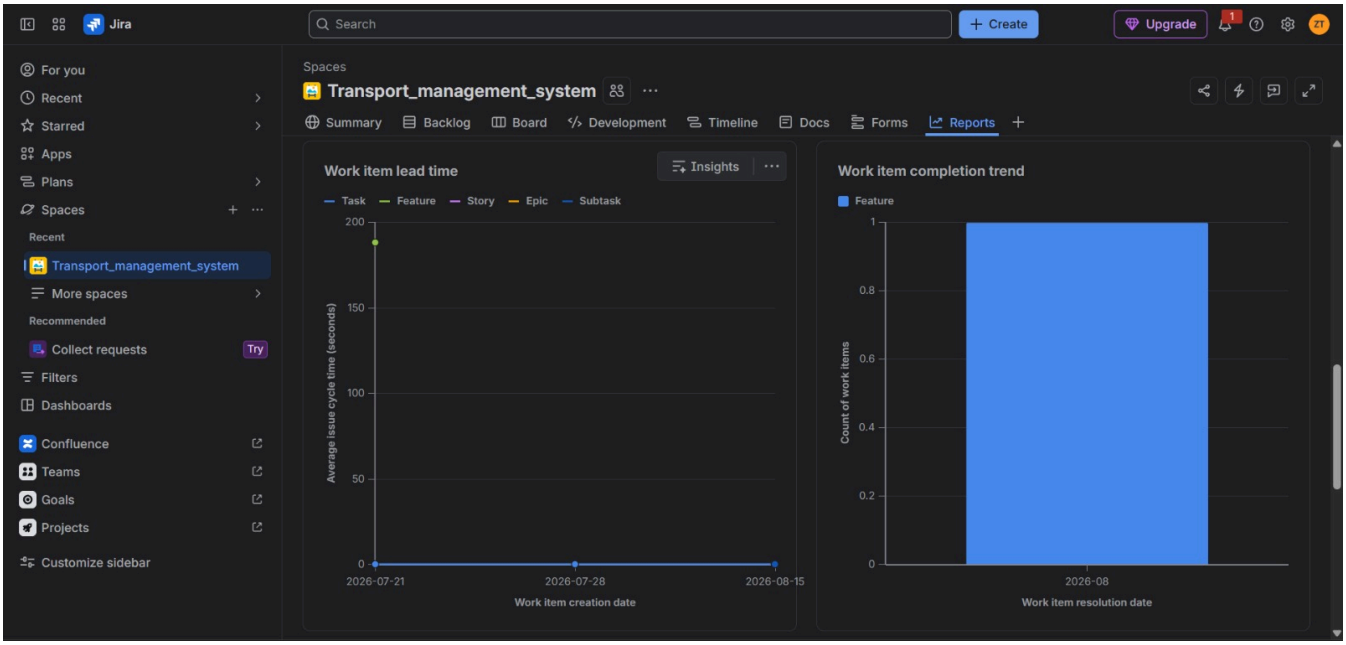
B.1 Code Screenshot:

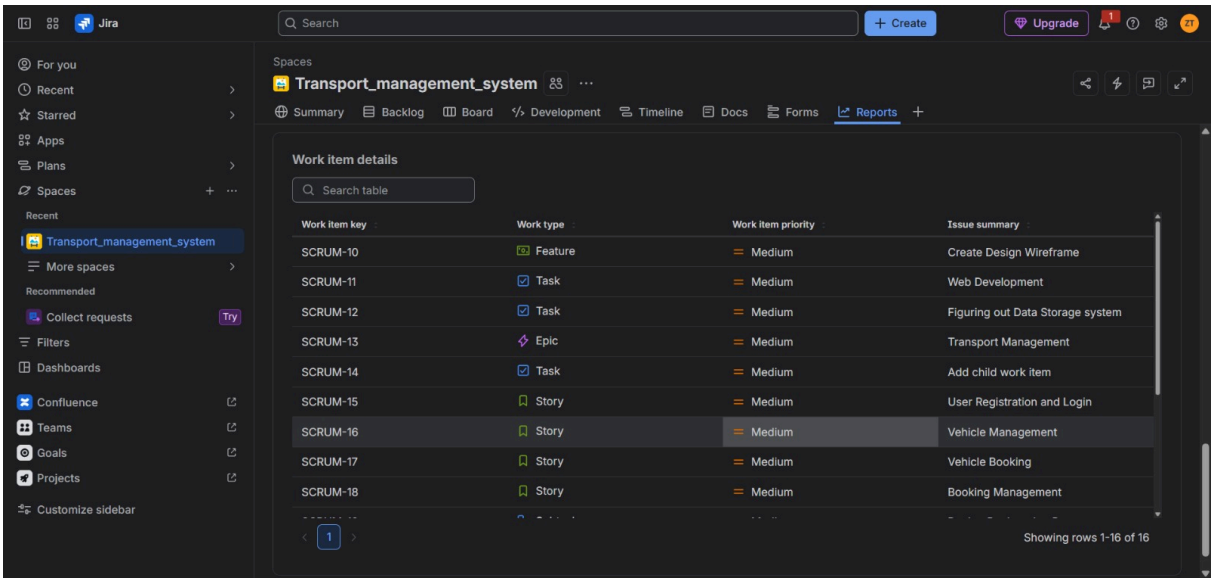


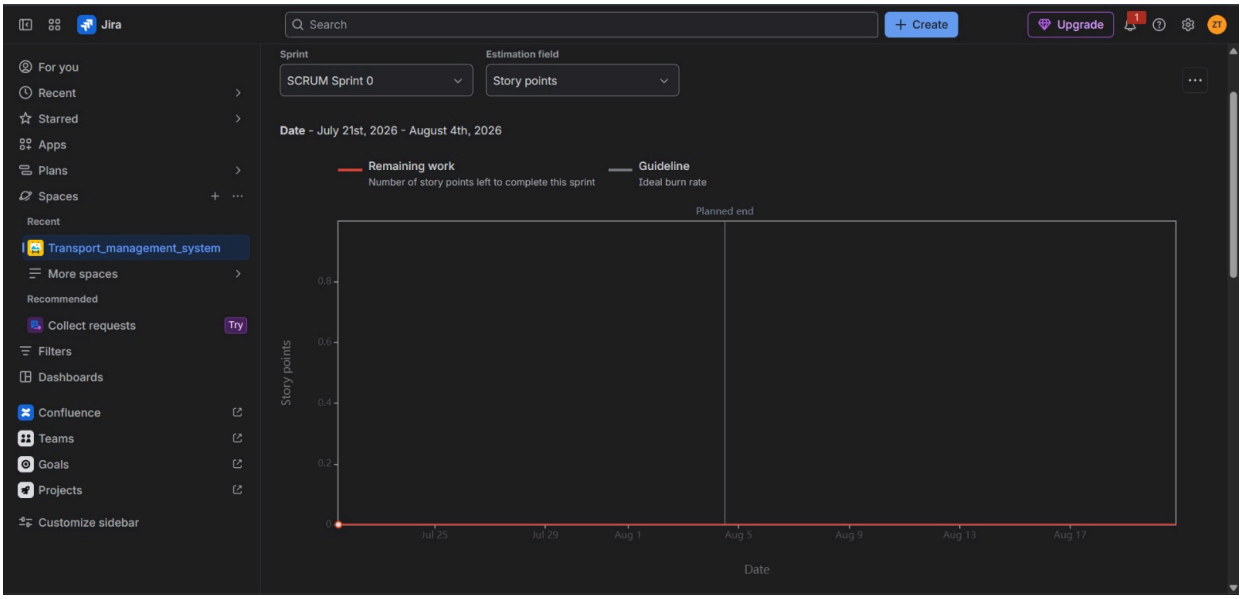


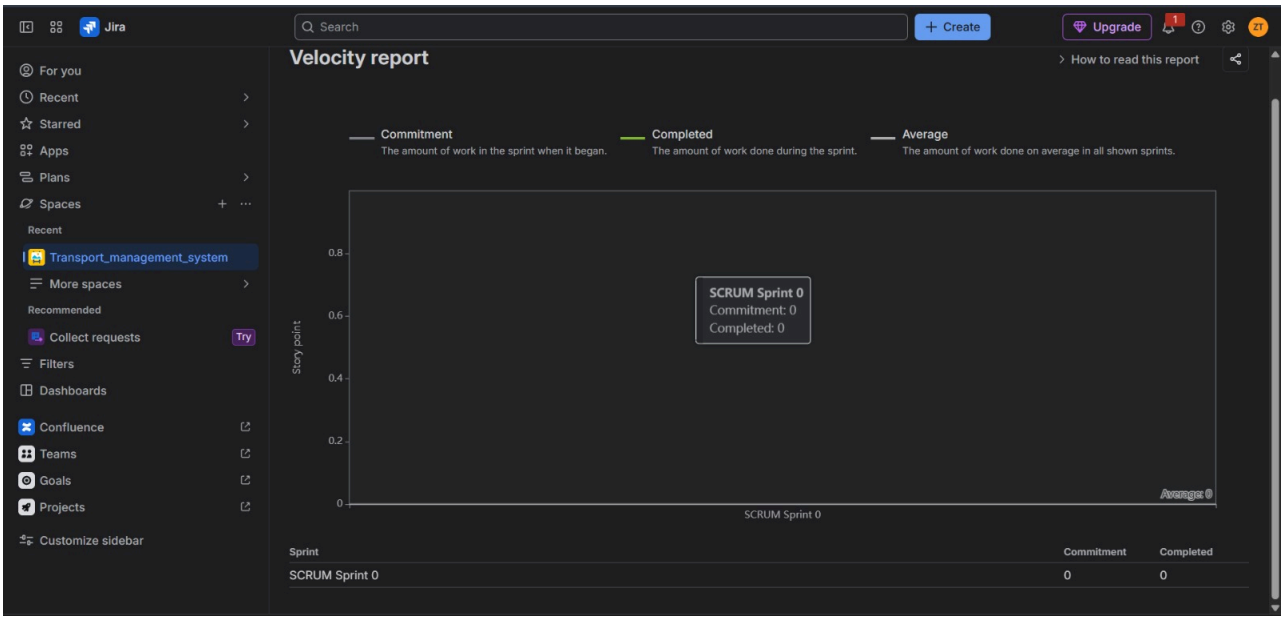

Jira
Search
Create
Upgrade
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Transport_management_system
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Confluence
Teams
Goals
Projects
Customize sidebar
Spaces
Transport_management_system
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Reports
More reports
Edit
0 work items
completed in the last 7 days
12 work items
updated in the last 7 days
9 work items
created in the last 7 days
1 work item
due in the next 7 days
Work items by status
16
Total value
To Do
In Progress
Done
Work items by type
Insights
16
Total value
Task
Story
Feature
Subtask
Epic
Work items by assignee
16
Total value
anuragamale2425
ZOLAGE PRAJWAL ...
yashmohite2425
adinathjagtap2425
Unassigned


Jira
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Confluence
Teams
Goals
Projects
Customize sidebar
Spaces
Transport_management_system
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Reports
Work item creation trend
Done
In Progress
To Do
Count of work items
8
6
4
2
0
2026-07-21
2026-07-28
2026-08-15
Work item creation date
Work item cycle time
Average issue cycle time (seconds)
188
2026-08-20
Work item completed date
Work item lead time
Insights
Task
Feature
Story
Epic
Subtask
200
Work item completion trend
Feature
1
0.8


Jira
Search
Create
Upgrade
For you
Recent
Starred
Apps
Plans
Spaces
Recent
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More spaces
Recommended
Collect requests
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Dashboards
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Goals
Projects
Customize sidebar
Spaces
Transport_management_system
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Reports
Work item lead time
Insights
Task
Feature
Story
Epic
Subtask
Average issue cycle time (seconds)
200
150
100
50
0
2026-07-21
2026-07-28
2026-08-15
Work item creation date
Work item completion trend
Feature
Count of work items
1
0.8
0.6
0.4
0.2
0
2026-08
Work item resolution date


Jira
Search
Create
Upgrade
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Transport_management_system
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Confluence
Teams
Goals
Projects
Customize sidebar
Spaces
Transport_management_system
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Reports
Work item details
Search table
Work item key
Work type
Work item priority
Issue summary
SCRUM-10 Feature Medium Create Design Wireframe
SCRUM-11 Task Medium Web Development
SCRUM-12 Task Medium Figuring out Data Storage system
SCRUM-13 Epic Medium Transport Management
SCRUM-14 Task Medium Add child work item
SCRUM-15 Story Medium User Registration and Login
SCRUM-16 Story Medium Vehicle Management
SCRUM-17 Story Medium Vehicle Booking
SCRUM-18 Story Medium Booking Management
1
Showing rows 1-16 of 16


Jira
Search
Create
Upgrade
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Transport_management_system
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Confluence
Teams
Goals
Projects
Customize sidebar
Sprint
SCRUM Sprint 0
Estimation field
Story points
Date - July 21st, 2026 - August 4th, 2026
Remaining work
Number of story points left to complete this sprint
Guideline
Ideal burn rate
Planned end
Story points
0.8
0.6
0.4
0.2
0
Jul 25
Jul 29
Aug 1
Aug 5
Aug 9
Aug 13
Aug 17
Date


Jira
Search
Create
Upgrade
Velocity report
How to read this report
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Transport_management_system
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Confluence
Teams
Goals
Projects
Customize sidebar
Commitment
The amount of work in the sprint when it began.
Completed
The amount of work done during the sprint.
Average
The amount of work done on average in all shown sprints.
Story point
0.8
0.6
0.4
0.2
0
SCRUM Sprint 0
Commitment: 0
Completed: 0
Average: 0
SCRUM Sprint 0
Sprint
Commitment
Completed
SCRUM Sprint 0
0
0

B.2 Input and Output:

Input

The following inputs were used to perform the experiment:

- **Project Name:** Transport_management_system
- **Project Type:** Scrum Project
- **Sprint Name:** SCRUM Sprint 0
- **Work Items:**
    - Requirements Analysis (SCRUM-9)
    - Create Design Wireframe (SCRUM-10)
- **Estimation Field:** Story Points
- **Workflow Columns:**
    - To Do
    - In Progress
    - In Review
    - Done

The work items were tracked throughout the spr\int by changing their status from **To Do** to **In Progress** and finally to **Done**.

Output

The following results were obtained:

1. The Scrum Board was successfully used to monitor the progress of work items during the sprint.
2. The work item **Create Design Wireframe (SCRUM-10)** was moved from **To Do** to **In Progress** and then to **Done**.
3. The progress of the sprint was visualized using the **Sprint Burndown Chart**.
4. The Burndown Chart displayed the remaining work based on the selected estimation field, **Story Points**.
5. The sprint was completed successfully.
6. The **Velocity Report** was generated after completing the sprint.
7. The Velocity Report showed the completed sprint, but the Commitment and Completed values were **0** because Story Points were not assigned to the work items.

Thus, the experiment successfully demonstrated the process of tracking sprint progress and accessing Agile reports in Jira.

B.3 Conclusion:

Thus, the Sprint progress was successfully tracked using Jira's Burndown and Velocity Charts. The experiment demonstrated how Agile teams can monitor Sprint work, analyze completed tasks, evaluate team performance, and improve planning for future Sprints.